PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Practice Midterm A - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Choose 3 of the following 4 problems. Complete every part of each chosen problem.

Write all rough calculations and final answers on separate paper. Show your working.

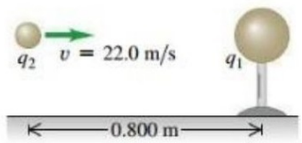
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.5

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80\ \mu C$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80\ \mu C$ and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2, is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1?

Figure **E23.5**



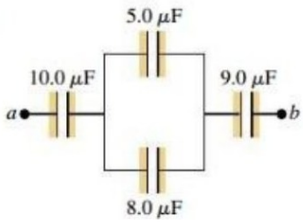
2. [____] Homework 4 - Question 22.19

22.19 •• A hollow, conducting sphere with an outer radius of 0.250 m and an inner radius of 0.200 m has a uniform surface charge density of $+6.37 \times 10^{-6}\ C/m^2$. A charge of $-0.500\ \mu C$ is now introduced at the center of the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

3. [____] Homework 7 - Question 24.14

24.14 • **Figure E24.14** shows a system of four capacitors, where the potential difference across *ab* is 50.0 V. (a) Find the equivalent capacitance of this system between *a* and *b*. (b) How much charge is stored by this combination of capacitors? (c) How much charge is stored in each of the 10.0-μF and the 9.0-μF capacitors?

Figure **E24.14**



4. [____] Tutorial 6 - Question 47

47. A parallel-plate capacitor in air has a plate separation of 1.50 cm and a plate area of 25.0 cm^2. The plates are charged to a potential difference of 250 V and disconnected from the source. The capacitor is then immersed in distilled water. Determine (a) the charge on the plates before and after immersion, (b) the capacitance and potential difference after immersion, and (c) the change in energy of the capacitor. Assume the liquid is an insulator.

PHYS 210 Electricity and Magnetism

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Practice Midterm B - CK #2 preparation

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Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.7

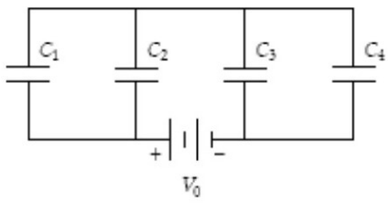
23.7 •• Two protons, starting several meters apart, are aimed directly at each other with speeds of 2.00×10^5 m/s, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

2. [____] Textbook 15e - Question 22.40

22.40 • **A Sphere in a Sphere.** A solid conducting sphere carrying charge q has radius a. It is inside a concentric hollow conducting sphere with inner radius b and outer radius c. The hollow sphere has no net charge. (a) Derive expressions for the electric-field magnitude in terms of the distance r from the center for the regions $r < a$, $a < r < b$, $b < r < c$, and $r > c$. (b) Graph the magnitude of the electric field as a function of r from $r = 0$ to $r = 2c$. (c) What is the charge on the inner surface of the hollow sphere? (d) On the outer surface? (e) Represent the charge of the small sphere by four plus signs. Sketch the field lines of the system within a spherical volume of radius $2c$.

3. [____] Tutorial 7 - Question 1

1. What is the total energy stored by C_3 when $C_1 = 50\ \mu$F, $C_2 = 30\ \mu$F, $C_3 = 36\ \mu$F, $C_4 = 12\ \mu$F, and $V_0 = 30$ V?



a. 6.3 mJ
b. 25 mJ
c. 57 mJ
d. 1.6 mJ
e. 14 mJ

4. [____] Tutorial 6 - Question 69

69. A parallel-plate capacitor of plate separation d is charged to a potential difference ΔV_0. A dielectric slab of thickness d and dielectric constant κ is introduced between the plates while the battery remains connected to the plates. (a) Show that the ratio of energy stored after the dielectric is introduced to the energy stored in the empty capacitor is $U/U_0 = \kappa$. Give a physical explanation for this increase in stored energy. (b) What happens to the charge on the capacitor? (Note that this situation is not the same as in Example 26.7, in which the battery was removed from the circuit before the dielectric was introduced.)

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Practice Midterm C - CK #2 preparation

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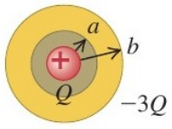
1. [____] Textbook 15e - Question 23.75

23.75 • An alpha particle with kinetic energy 9.50 MeV (when far away) collides head-on with a lead nucleus at rest. What is the distance of closest approach of the two particles? (Assume that the lead nucleus remains stationary and may be treated as a point charge. The atomic number of lead is 82. The alpha particle is a helium nucleus, with atomic number 2.)

2. [____] Textbook 15e - Question 22.42

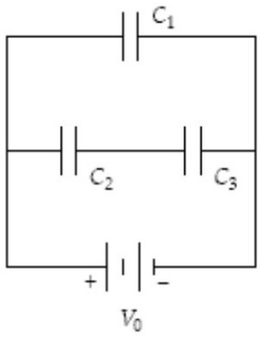
22.42 • A conducting spherical shell with inner radius a and outer radius b has a positive point charge Q located at its center. The total charge on the shell is $-3Q$, and it is insulated from its surroundings (**Fig. P22.42**). (a) Derive expressions for the electric-field magnitude E in terms of the distance r from the center for the regions $r < a$, $a < r < b$, and $r > b$. What is the surface charge density (b) on the inner surface of the conducting shell; (c) on the outer surface of the conducting shell? (d) Sketch the electric field lines and the location of all charges. (e) Graph E as a function of r.

Figure **P22.42**



3. [____] Tutorial 7 - Question 2

2. What is the potential difference across C_2 when $C_1 = 5.0\ \mu$F, $C_2 = 15\ \mu$F, $C_3 = 30\ \mu$F, and $V_0 = 24$ V?



a. 21 V
b. 19 V
c. 16 V
d. 24 V
e. 8.0 V

4. [____] Tutorial 7 - Question 7

7. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. It is then disconnected from the battery and the space between the plates is filled with a material of dielectric constant 3. After the dielectric is added, the magnitudes of the charge on the plates and the potential difference between them are
   a. $\frac{1}{3}Q_0, \frac{1}{3}V_0$.
   b. $Q_0, \frac{1}{3}V_0$.
   c. Q_0, V_0.
   d. $Q_0, 3V_0$.
   e. $3Q_0, 3V_0$.

PHYS 210 Electricity and Magnetism

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Practice Midterm D - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

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Write all rough calculations and final answers on separate paper. Show your working.

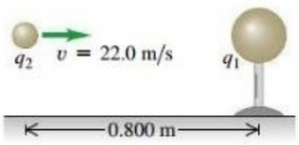
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1. [____] Homework 5 - Question 23.5

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80\ \mu C$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80\ \mu C$ and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2, is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1?

Figure **E23.5**



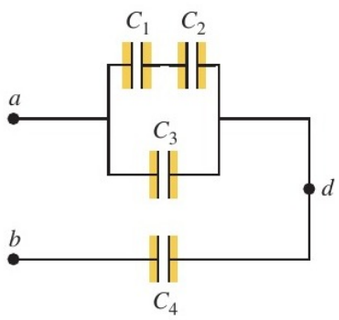
2. [____] Tutorial 4 - Question 7

7. A 4.0-pC point charge is placed at the center of a hollow (inner radius = 2.0 cm, outer radius = 4.0 cm) conducting sphere which has a net charge of 4.0 pC. Determine the magnitude of the electric field at a point which is 6.0 cm from the point charge.
   a. 35 N/C
   b. 25 N/C
   c. 30 N/C
   d. 20 N/C
   e. 10 N/C

3. [____] Textbook 15e - Question 24.17

24.17 • In **Fig. E24.17**, each capacitor has $C = 4.00\ \mu F$ and $V_{ab} = +28.0$ V. Calculate (a) the charge on each capacitor; (b) the potential difference across each capacitor; (c) the potential difference between points a and d.

Figure **E24.17**



4. [____] Textbook 15e - Question 24.31

24.31 • A 12.5 μF capacitor is connected to a power supply that keeps a constant potential difference of 24.0 V across the plates. A piece of material having a dielectric constant of 3.75 is placed between the plates, completely filling the space between them. (a) How much energy is stored in the capacitor before and after the dielectric is inserted? (b) By how much did the energy change during the insertion? Did it increase or decrease?

PHYS 210 Electricity and Magnetism

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Practice Midterm E - CK #2 preparation

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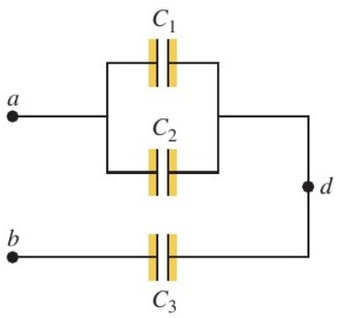
2. [____] Tutorial 4 - Question 9

9. A point charge of 6.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm, outer radius = 2.0 cm) which has a net charge of –4.0 nC. Determine the resulting charge density on the inner surface of the conducting sphere.
   a. +4.8 μC/m^2
   b. –4.8 μC/m^2
   c. –9.5 μC/m^2
   d. +9.5 μC/m^2
   e. –8.0 μC/m^2

3. [____] Textbook 15e - Question 24.20

24.20 • In **Fig. E24.20**, $C_1 = 6.00\ \mu$F, $C_2 = 3.00\ \mu$F, and $C_3 = 5.00\ \mu$F. The capacitor network is connected to an applied potential V_{ab}. After the charges on the capacitors have reached their final values, the charge on C_2 is 30.0 μC. (a) What are the charges on capacitors C_1 and C_3? (b) What is the applied voltage V_{ab}?

Figure **E24.20**



4. [____] Textbook 15e - Question 24.39

24.39 • A constant potential difference of 12 V is maintained between the terminals of a 0.25 μF, parallel-plate, air capacitor. (a) A sheet of Mylar is inserted between the plates of the capacitor, completely filling the space between the plates. When this is done, how much additional charge flows onto the positive plate of the capacitor (see Table 24.1)? (b) What is the total induced charge on either face of the Mylar sheet? (c) What effect does the Mylar sheet have on the electric field between the plates? Explain how you can reconcile this with the increase in charge on the plates, which acts to *increase* the electric field.

PHYS 210 Electricity and Magnetism

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Practice Midterm F - CK #2 preparation

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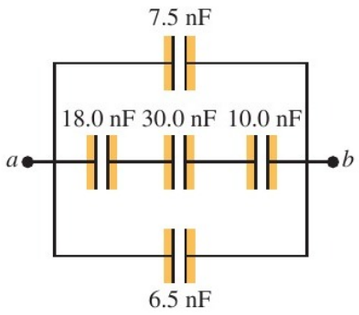
2. [____] Tutorial 4 - Question 8

8. The axis of a long hollow metallic cylinder (inner radius = 1.0 cm, outer radius = 2.0 cm) coincides with a long wire. The wire has a linear charge density of −8.0 pC/m, and the cylinder has a net charge per unit length of −4.0 pC/m. Determine the magnitude of the electric field 3.0 cm from the axis.
   a. 5.4 N/C
   b. 7.2 N/C
   c. 4.3 N/C
   d. 3.6 N/C
   e. 2.4 N/C

3. [____] Textbook 15e - Question 24.21

24.21 •• For the system of capacitors shown in **Fig. E24.21**, a potential difference of 25 V is maintained across *ab*. (a) What is the equivalent capacitance of this system between *a* and *b*? (b) How much charge is stored by this system? (c) How much charge does the 6.5 nF capacitor store? (d) What is the potential difference across the 7.5 nF capacitor?

Figure **E24.21**



4. [____] Textbook 15e - Question 24.42

24.42 • A parallel-plate capacitor has capacitance $C = 12.5$ pF when the volume between the plates is filled with air. The plates are circular, with radius 3.00 cm. The capacitor is connected to a battery, and a charge of magnitude 25.0 pC goes onto each plate. With the capacitor still connected to the battery, a slab of dielectric is inserted between the plates, completely filling the space between the plates. After the dielectric has been inserted, the charge on each plate has magnitude 45.0 pC. (a) What is the dielectric constant K of the dielectric? (b) What is the potential difference between the plates before and after the dielectric has been inserted? (c) What is the electric field at a point midway between the plates before and after the dielectric has been inserted?

PHYS 210 Electricity and Magnetism

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Practice Midterm G - CK #2 preparation

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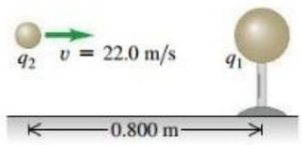
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Figure **E23.5**



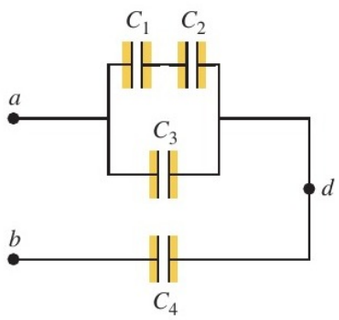
2. [____] Textbook 15e - Question 22.25

22.25 • A conductor with an inner cavity, like that shown in Fig. 22.23c, carries a total charge of +5.00 nC. The charge within the cavity, insulated from the conductor, is −6.00 nC. How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

3. [____] Textbook 15e - Question 24.52

24.52 •• In Fig. E24.17, $C_1 = 6.00\ \mu F$, $C_2 = 3.00\ \mu F$, $C_3 = 4.00\ \mu F$, and $C_4 = 8.00\ \mu F$. The capacitor network is connected to an applied potential difference V_{ab}. After the charges on the capacitors have reached their final values, the voltage across C_3 is 40.0 V. What are (a) the voltages across C_1 and C_2, (b) the voltage across C_4, and (c) the voltage V_{ab} applied to the network?

Figure **E24.17**



4. [____] Textbook 15e - Question 24.44

24.44 • A parallel-plate capacitor has plates with area 0.0225 m^2 separated by 1.00 mm of Teflon. (a) Calculate the charge on the plates when they are charged to a potential difference of 12.0 V. (b) Use Gauss's law [Eq. (24.23)] to calculate the electric field inside the Teflon. (c) Use Gauss's law to calculate the electric field if the voltage source is disconnected and the Teflon is removed.

PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Practice Midterm H - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

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1. [____] Homework 5 - Question 23.7

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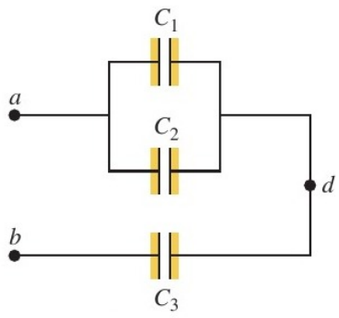
2. [____] Tutorial 4 - Question 6

6. Charge of uniform density (40 pC/m^2) is distributed on a spherical surface (radius = 1.0 cm), and a second concentric spherical surface (radius = 3.0 cm) carries a uniform charge density of 60 pC/m^2. What is the magnitude of the electric field at a point 4.0 cm from the center of the two surfaces?
   a. 3.8 N/C
   b. 4.1 N/C
   c. 3.5 N/C
   d. 3.2 N/C
   e. 0.28 N/C

3. [____] Textbook 15e - Question 24.55

24.55 •• In Fig. E24.20, $C_1 = 3.00\ \mu$F and $V_{ab} = 150$ V. The charge on capacitor C_1 is 150 μC and the charge on C_3 is 450 μC. What are the values of the capacitances of C_2 and C_3?

Figure **E24.20**



4. [____] Textbook 15e - Question VP24.11.1

VP24.11.1 The two plates of a parallel-plate capacitor each have area 0.460 m^2, are 3.00 mm apart, and initially have vacuum between them. A power supply is attached to the capacitor, charging it to 4.00 kV, and is then disconnected. A dielectric sheet is then inserted that fills the space between the plates. The potential difference between the plates decreases to 2.50 kV, and the charge on each plate remains constant. Find (a) the original capacitance, (b) the charge on the capacitor plates, (c) the dielectric constant of the dielectric, (d) the new capacitance after you insert the dielectric, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.